Thermal Expansion

Exam code: 0625 & 0972

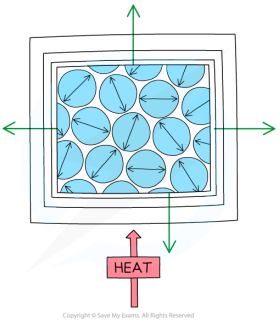
Download PDF

Written by: Lindsay Gilmour
Reviewed by: Caroline Carroll
Updated on 3 January 2025

Thermal expansion

- When a material is heated at constant pressure:
  - Its temperature increases
  - Its overall volume increases (it **expands**)
  - Its density decreases
- This expansion happens because:
  - The molecules start to move around (or vibrate) faster as they gain kinetic energy
  - This causes them to collide with each other more often and push each other apart

Thermal expansion diagram



When a solid is heated, the molecules vibrate more, pushing each other apart

Thermal expansion in terms of particles

Extended tier only

- Thermal expansion occurs in solids, liquids and gases
- When temperature is increased (at constant pressure):
  - Solids will tend to expand the least
  - Gases expand the most
  - Liquids fall in between the two
- This behaviour is influenced by the distances and strength of the forces between particles in each state

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

State	Magnitude of Expansion	Explanation
Solid	Expands slightly	The low energy molecules cannot overcome the intermolecular forces of attraction holding them together
Liquid	Expands more than solids	The molecules have enough energy to partially overcome the intermolecular forces of attraction holding them together
Gas	Expand significantly	The high energy molecules have enough energy to completely overcome the intermolecular forces of attraction holding them together

Uses & consequences of thermal expansion

- The thermal expansion of materials can have some **useful applications** as well as some **undesirable consequences**

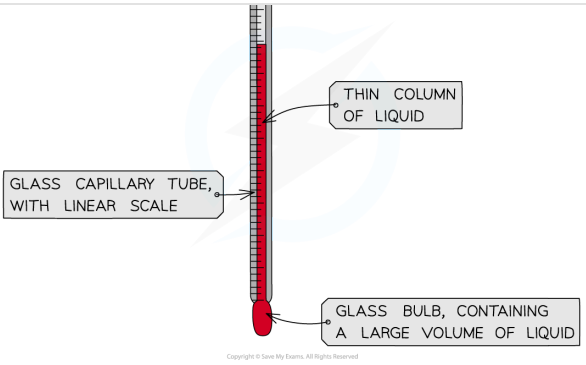
Applications of thermal expansion

- Useful applications of thermal expansion include:
  - Liquid-in-glass thermometers
  - Temperature-activated switches

Liquid-in-glass thermometer

- A liquid-in-glass thermometer relies on the expansion of liquids to measure temperature
- It consists of a thin glass capillary tube containing a liquid that expands with temperature
  - At one end of the tube is a glass bulb, containing a large volume of the liquid which expands into the narrow tube when heated
  - A scale along the side of the tube allows the temperature to be measured based on the length of liquid within the tube

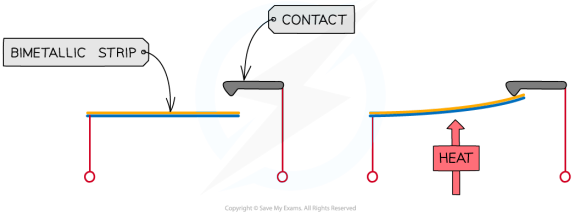
Liquid-in-glass thermometer



As the bulb is heated, the liquid expands and moves along the capillary tube

Temperature-activated switches

- Temperature-activated switches utilise a **bimetallic** (made from two types of metal) strip
- It consists of two metals that expand at different rates and bends by a predictable amount at a given temperature



The bimetallic strip will bend upwards when heated, closing the circuit

Consequences of thermal expansion

- The expansion of solid materials can cause them to buckle if they get too hot
- This could include:
  - Metal railway tracks
  - Road surfaces
  - Bridges

- Objects that are prone to buckling in this way have gaps built in to creates space for the expansion to happen without causing damage

Examiner Tips and Tricks

Remember that it is **the material that expands, not the molecules**. This trips up many students, losing marks.

As heat is added;

4. And move apart
   - This separation of the the molecules makes the substance bigger!

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder →

Test yourself Flashcards

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.